A First Course in Abstract Algebra

Section 38: Free Abelian Groups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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38 Free Abelian Groups

Exercise 38.1: A basis of $\mathbb{Z}^3$ with no zero coordinates

Take $u = (1, 1, 1)$, $v = (1, 2, 2)$, $w = (1, 2, 3)$. For any $(e, f, g) \in \mathbb{Z}^3$, solving $xu + yv + zw = (e, f, g)$ by subtracting coordinates gives

$$z = g - f, \qquad y = 2f - e - g, \qquad x = 2e - f.$$

These are integers and substitution recovers all three coordinates. The formulas also give a unique coefficient triple, so the vectors generate $\mathbb{Z}^3$ and are independent over $\mathbb{Z}$. Thus they form a basis, with every coordinate nonzero.

Exercise 38.2: The vectors $(2, 1)$ and $(3, 1)$

Yes. For an arbitrary (e, f), the equations $2x + 3y = e$, $x + y = f$ give

$$y = e - 2f, \qquad x = 3f - e.$$

Both coefficients are integers, so every integer pair is generated. They are uniquely determined, so the only representation of zero is $x = y = 0$. These are precisely the generating and independence requirements for a basis of a free abelian group.

Exercise 38.3: The vectors $(2, 1)$ and $(4, 1)$

No. Any integer combination $x(2, 1) + y(4, 1) = (2x + 4y, x + y)$ has even first coordinate. Therefore $(1, 0)$ is not generated. The two vectors are linearly independent over $\mathbb{R}$ (their determinant is -2), but that does not make them a $\mathbb{Z}$-basis: integer coefficients, rather than arbitrary real coefficients, are required.

Exercise 38.4: The determinant criterion for an integer basis

Put $D = ad - bc$, the determinant of the matrix with columns (a, b) and (c, d). If $D = \pm 1$, solving $x(a, b) + y(c, d) = (e, f)$ gives

$$x = \frac{de - cf}{D}, \qquad y = \frac{af - be}{D}.$$

These coefficients are integers and unique for all $e, f \in \mathbb{Z}$, proving the basis property. Conversely, if the two vectors are a basis, there is an integer matrix B whose columns express $(1, 0), (0, 1)$ in that basis. For $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ this gives $AB = I$. Taking determinants yields $D \det B = 1$. Both factors are integers, so $D = \pm 1$. Hence the necessary and sufficient condition is $ad - bc = \pm 1$.

Exercise 38.5: Definition of rank

Replace “a generating set” by “a basis.” The rank is the cardinality of any basis of the free abelian group. Generating sets can contain redundant elements: $\{1\}$ and $\{1, 2\}$ both generate $\mathbb{Z}$ but have different sizes. Bases have invariant cardinality.

Exercise 38.6: Definition of an abelian basis

The proposed definition is correct. The zero relation with distinct basis elements forces every integer coefficient to vanish, and the set generates the group. The reverse implication in the independence condition is automatic, since a sum with all coefficients zero is zero. A group having such a basis is free abelian.

Exercise 38.7: A proper subgroup of the same rank

For any positive integer r, take $G = \mathbb{Z}^r$ and $K = 2\mathbb{Z} \times \mathbb{Z}^{r-1}$. It is proper since $(1, 0, \ldots, 0) \notin K$. The r vectors $2e_1, e_2, \ldots, e_r$ generate K and are independent over $\mathbb{Z}$, so K has rank r, just as G does. In particular already $2\mathbb{Z} < \mathbb{Z}$ gives a rank-1 example.

Exercise 38.8: True or false: free abelian groups

(a) True. If a multiple of an element is zero, uniqueness of basis coefficients makes every coefficient zero.

(b) True by the structure theorem: the torsion cyclic factors vanish, leaving $\mathbb{Z}^r$.

(c) True: $\mathbb{Z}^r$ has the standard r-element basis.

(d) True. If G has Betti number r and is generated by r elements, write $G \cong \mathbb{Z}^r \oplus T$. A prime $p \mid |T|$ would give $G/pG \cong \mathbb{F}_p^r \oplus T/pT$ of dimension greater than r, impossible with r generators. Thus $T = 0$.

(e) True. The larger set still contains every generator in X, so still generates all of G.

(f) False. $\{1\}$ is a basis of $\mathbb{Z}$, but $\{1, 2\}$ is dependent: $2 \cdot 1 - 1 \cdot 2 = 0$.

(g) False. $\mathbb{Z}$ has only two bases, $\{1\}$ and $\{-1\}$.

(h) True. If $x_1, x_2, \ldots$ is a basis, replacing x_1 by $x_1 + kx_2$ gives a different basis for each $k \in \mathbb{Z}$, since the old x_1 is recovered by subtracting kx_2.

(i) True: a subgroup of $\mathbb{Z}^r$ is free abelian, as proved in this section.

(j) False. For $G = \mathbb{Z}$, $K = 2\mathbb{Z}$, the quotient is $\mathbb{Z}_2$, which has nonzero torsion and is not free abelian.

Exercise 38.9: The coordinate isomorphism

For a basis $x_1, \ldots, x_r$, define $\phi(\sum n_i x_i) = (n_1, \ldots, n_r)$. Uniqueness of a basis expansion makes this well defined, including the zero vector. If $a = \sum n_i x_i$ and $b = \sum m_i x_i$, then $a + b = \sum (n_i + m_i) x_i$, so $\phi(a + b) = \phi(a) + \phi(b)$. If $\phi(a) = 0$, all coefficients are zero, hence $a = 0$; thus the map is injective. Given any integer tuple $(n_1, \ldots, n_r)$, its preimage is $\sum n_i x_i$, so the map is onto. This completes the claimed isomorphism $G \cong \mathbb{Z}^r$.

Exercise 38.10: A free abelian group has no nonzero torsion

Let $a = \sum_{i=1}^{r} n_i x_i$ in a basis, with finitely many coefficients involved, and suppose $ma = 0$ for an integer $m \neq 0$. Then $\sum_i (mn_i)x_i = 0$. Independence forces $mn_i = 0$ in $\mathbb{Z}$ for each i. Since $m \neq 0$, every $n_i = 0$, so $a = 0$. Thus a nonzero element cannot have finite order; the argument works even for an infinite basis because each expansion is finite.

Exercise 38.11: Products of free abelian groups

If X is a basis of G and Y a basis of G', use

$$\{(x, 0) : x \in X\} \ \cup \ \{(0, y) : y \in Y\}.$$

It generates $G \times G'$ by expanding each coordinate separately. A zero integer combination has zero first coordinate, forcing all X-coefficients to be zero, and zero second coordinate, forcing all Y-coefficients to be zero. Thus the displayed set is a basis, and the product is free abelian. This is a finite product of groups; even when the bases are infinite, each individual expansion remains finite.

Exercise 38.12: Finitely generated torsion-free groups

A free abelian group of finite rank has a finite basis and hence is finitely generated; Exercise 38.10 proves it is torsion free. Conversely the structure theorem writes a finitely generated abelian group as $\mathbb{Z}^r \oplus T$, where T is finite. If there is no nonzero element of finite order, T must be trivial. Thus the group is $\mathbb{Z}^r$, which has its standard basis. The zero group is included by the usual empty-basis/rank-zero convention; if "free abelian" is restricted to nonzero groups as in the section's initial definition, state the equivalence for nonzero groups.

Exercise 38.13: Why $\mathbb{Q}$ is not free abelian

Two nonzero rational numbers $x = a/b$ and $y = c/d$ satisfy $(bc)x - (ad)y = 0$, with both coefficients nonzero when $a, c \neq 0$. Thus no independent set over $\mathbb{Z}$ can contain two distinct nonzero rational numbers; zero cannot be a basis element either. A basis for the nonzero group $\mathbb{Q}$ would therefore contain just one nonzero rational q. But its generated subgroup consists of integer multiples of q, and $q/2$ is not one of them. Hence one element cannot generate $\mathbb{Q}$, so $\mathbb{Q}$ is not free abelian.

Exercise 38.14: The p-primary torsion subgroup

The set T_p contains zero. If $p^a x = 0$ and $p^b y = 0$, let $c = \max(a, b)$; then $p^c(x - y) = 0$. Hence $x - y$ has order a power of p (or is zero), and lies in T_p. The one-step subgroup criterion proves $T_p \leq T$. The commutativity of T is used when distributing multiplication by p^c over a difference.

Exercise 38.15: Locating the p-primary factors

In a prime-power product decomposition of T, an element is killed by a power of p exactly when all its coordinates in q-power factors with $q \neq p$ vanish. Indeed, multiplication by p is invertible on a cyclic q-group, because $\gcd(p, q) = 1$. Its repeated powers therefore cannot kill a nonzero coordinate.

Conversely the remaining p-power coordinates are all killed by some common power of p. Restricting the product isomorphism to T_p thus identifies it exactly with the product of the cyclic p-power factors.

Exercise 38.16: Elements killed by an integer

The multiplication map $\mu_n : G \to G$, $x \mapsto nx$, is a homomorphism because G is abelian. Therefore $G[n] = \ker \mu_n$ is a subgroup. Directly, $n0 = 0$, and if $nx = ny = 0$, then $n(x - y) = nx - ny = 0$, proving the same fact without invoking kernels. For a general nonabelian group the power map need not be a homomorphism; the abelian assumption is essential.

Exercise 38.17: The p-torsion of a cyclic p-group

In $\mathbb{Z}_{p^r}$ the equation $px = 0$ means $p^r \mid px$, equivalently $p^{r-1} \mid x$. Thus

$$\mathbb{Z}_{p^r}[p] = \{0, p^{r-1}, 2p^{r-1}, \ldots, (p-1)p^{r-1}\}.$$

It is cyclic generated by p^{r-1} and has order p, so the map $\mathbb{Z}_p \to \mathbb{Z}_{p^r}[p]$ sending k to kp^{r-1} is a well-defined bijective homomorphism.

Exercise 38.18: Counting p-torsion in a product

An element of a direct product is killed by p exactly when every coordinate is killed by p. Hence

$$\left(\prod_{i=1}^{m} \mathbb{Z}_{p^{r_i}}\right)[p] = \prod_{i=1}^{m} (\mathbb{Z}_{p^{r_i}}[p]) \cong \mathbb{Z}_p^m$$

by Exercise 38.17. In particular its size is p^m; this size is intrinsic to the group, independent of a chosen decomposition.

Exercise 38.19: Uniqueness of the p-power exponents

(a) Apply the intrinsic construction $T_p[p]$ to the two decompositions. Their orders are p^m and p^n, so $m = n$. **(b)** Arrange exponents increasingly. Suppose that $r_i = s_i$ for $i < j$ and, seeking a contradiction, $r_j < s_j$. The group $p^{r_j}T_p$ has from the first decomposition one nonzero cyclic factor for each $r_i > r_j$, at most $m - j$ factors. From the second decomposition every index $i \geq j$ has $s_i \geq s_j > r_j$, giving at least $m - j + 1$ nonzero factors; the indices before j have the same behavior in both decompositions. Part (a), applied to this intrinsic subgroup, forbids different numbers of nonzero p-power cyclic factors. Thus $r_j < s_j$ is impossible. Interchange the decompositions to rule out $s_j < r_j$. Start at $j = 1$ and induct to obtain $r_i = s_i$ for all i.

Exercise 38.20: From invariant factors to prime powers

Factor each torsion coefficient m_i into pairwise coprime prime powers. The Chinese remainder isomorphism gives

$$\mathbb{Z}_{m_i} \cong \prod_{p \mid m_i} \mathbb{Z}_{p^{v_p(m_i)}}.$$

Taking the product over i and regrouping by prime produces the prime-power decomposition of T. Its uniqueness follows from Exercises 38.14–19. Factors of order 1 are omitted, since they contribute only a trivial coordinate.

Exercise 38.21: Recovering the largest torsion coefficient

Because $m_i \mid m_{i+1}$, the valuations $v_p(m_i)$ are nondecreasing for each prime p. The largest exponent for p in the entire prime-power decomposition is consequently $v_p(m_r)$. The same intrinsic largest exponent is $v_p(n_s)$ in the other decomposition. Therefore, writing it as h_p,

$$m_r = \prod_{p \| T|} p^{h_p} = n_s.$$

All primes dividing T occur in this product, so equality of the prime valuations proves equality of the integers.

Exercise 38.22: Recovering all torsion coefficients

For each prime, list its exponents in the unique prime-power decomposition in increasing order. Pad each prime's list on the *left* with zeros to a common length. The last torsion coefficient is the product of the prime powers in the last column; the preceding coefficient is the product in the next-to-last column, and so on. This reconstructs m_{r-1} and n_{s-1} identically: for each prime take its second-largest exponent, or zero if it appears only once. Continue using third-largest exponents and so forth. Divisibility forces precisely this alignment of the lists, because valuations are nondecreasing. The number of columns is the largest number of nonzero exponents in any prime's list: otherwise the first coefficient would be 1, contrary to $m_1, n_1 > 1$. Thus the common length is both r and s, and every coefficient agrees. This proves $r = s$ and $m_i = n_i$ for all i.